AI-POWERED INTELLIGENT TALENT ACQUISITION SYSTEM

INDIVIDUAL THESIS

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Job–Candidate Matching & Explainable AI Engine

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ABSTRACT

Recruitment Involves Evaluating Candidate Qualifications Against Job Requirements To Identify Suitable Individuals For Available Positions. However, Manually Reviewing Large Numbers Of Resumes Can Be Time-Consuming, While Conventional Keyword-Based Screening May Overlook Relevant Skills And Qualifications Expressed Using Different Terminology. Furthermore, Matching Systems That Provide Scores Without Clear Explanations May Limit Recruiters' Ability To Understand Candidate Recommendations. These Challenges Highlight The Need For An Intelligent And Interpretable Job–Candidate Matching Approach.

This Research Focuses On The Job–Candidate Matching And Explainable Artificial Intelligence (Xai) Engine, An Individual Component Of The Ai-Powered Intelligent Talent Acquisition System. The Primary Objective Is To Compare Candidate Profiles With Job Descriptions, Assess Their Relevance, Rank Candidates According To Matching Criteria, And Provide Understandable Explanations To Support Recruitment Decision-Making. The Proposed Technical Approach Considers Natural Language Processing (Nlp), Text Preprocessing, Numerical Text Representation, Similarity Measurement, And Matching Evidence To Identify Relevant Qualifications And Potential Gaps Between Candidate Profiles And Job Requirements. Explainability Is Considered An Essential Aspect Of The Component, Enabling Recruiters To Examine The Evidence Underlying Matching Recommendations.

The Research Considers Matching Relevance, Candidate Ranking Quality, Interpretability, And The Limitations Of Automated Candidate Evaluation. Appropriate Evaluation Methods Are Required To Assess The Effectiveness And Reliability Of The Implemented Component. The Individual Contribution Is The Design And Development Of A Job–Candidate Matching And Ranking Component Intended To Support Structured Recruitment Screening Through Interpretable

Matching Results. Final Conclusions Regarding Its Effectiveness Will Be Based On Verified Implementation Evidence And Empirical Evaluation Findings.

Keywords: Job–Candidate Matching, Natural Language Processing, Explainable Artificial Intelligence, Candidate Ranking, Recruitment Decision Support

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I would also like to thank my project team members for their cooperation and coordination in developing the AI-Powered Intelligent Talent Acquisition System. Their collaborative efforts have contributed to the overall project, while my individual research focuses specifically on the Job–Candidate Matching and Explainable AI Engine.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
CV	Curriculum Vitae
IDF	Inverse Document Frequency
IR	Information Retrieval
JSON	JavaScript Object Notation
ML	Machine Learning
NLP	Natural Language Processing
REST	Representational State Transfer
TF-IDF	Term Frequency–Inverse Document Frequency
UI	User Interface
UX	User Experience

1 INTRODUCTION

1.1 Overall project context

Recruitment has largely been based on evaluating whether the resume of the candidate is consistent with the job requirements. However, the consistency with a job and the likelihood of the candidate to stay in the organization are two different dimensions in recruitment. Even if a candidate possesses all the required qualifications, he or she might be at risk of leaving due to certain issues like remuneration, work environment, career perspectives, relationship with the managers, or other factors related to the workplace. In the context of Sri Lanka, predictors of turnover include factors such as job satisfaction, organizational support, job stress, alternatives, and overall commitment.

The audit of the repository shows that the component makes use of a polyglot containerized microservices architecture. Specifically, the frontend built in React and Vite sends the requests for predictions through an Express API Gateway to the Attrition Orchestrator. The orchestrator interacts with services of a production CatBoost model, an early-attrition Logistic Regression model, and an explainability service. Under the current configuration, the attrition capability is decoupled from the matching capability while making use of its output for predictions.

1.2 Individual component context

The Job–Candidate Matching and Explainable AI Engine is designed to address the candidate screening dimension of the recruitment process. The component focuses on assessing the relevance of candidate profiles to specific job descriptions and ranking candidates according to their matching results. Unlike conventional screening approaches that rely primarily on manual review or basic keyword matching, the proposed component aims to provide a systematic approach to comparing candidate skills, qualifications, and experience against job requirements.

The Explainable AI functionality provides an additional dimension to the matching process by helping recruiters understand why a candidate receives a particular matching score or ranking. By highlighting relevant candidate attributes and potential

gaps in relation to the job requirements, the component aims to improve the transparency and interpretability of recruitment recommendations. This enables recruiters to review the matching results more effectively rather than relying solely on numerical scores when shortlisting candidates.

My individual contribution to the project focuses on researching job–candidate matching approaches, analysing candidate profiles and job descriptions, designing and implementing the matching mechanism, developing explanations for matching outcomes, evaluating candidate ranking performance, testing the component, and integrating it with the broader AI-Powered Intelligent Talent Acquisition System.

1.3 Problem statement

Recruiters often face difficulties in identifying suitable candidates from a large number of CVs. Manually comparing candidate skills, qualifications, and experience with job requirements requires considerable time and effort. Basic keyword-matching approaches may also overlook relevant candidates when similar skills or experiences are described using different terms.

Furthermore, candidate ranking alone does not explain why a particular candidate receives a higher matching score than others. Therefore, a job–candidate matching solution is needed to assess candidate relevance systematically while providing clear explanations of matching results. This research focuses on addressing these challenges through a matching and explainability component that supports recruiters in making more informed shortlisting decisions.

1.4 Research gap

Existing job–candidate matching approaches use techniques such as keyword matching and text similarity to compare candidate profiles with job descriptions. However, these approaches may not fully identify relevant candidates when skills and experience are expressed using different terminology.

Furthermore, some matching approaches provide candidate scores or rankings without clearly explaining the reasons behind the results. This creates a need for a solution that

combines candidate matching, ranking, and explainability to help recruiters understand how candidate profiles relate to job requirements. This research focuses on addressing this need through the development and evaluation of a Job–Candidate Matching and Explainable AI Engine.

1.5 Aim and objectives

The main aim of this research is to develop a Job–Candidate Matching and Explainable AI Engine that evaluates candidate profiles against job requirements, ranks candidates according to their relevance, and provides explanations to help recruiters understand the matching results.

The objectives are to investigate existing matching techniques, identify relevant information from candidate profiles and job descriptions, develop a candidate matching and ranking mechanism, provide understandable explanations of matching outcomes, and evaluate the effectiveness of the proposed component using appropriate evaluation methods.

1.6 Research questions or hypotheses

This research is guided by the following questions:

1. How can candidate profiles be matched with job descriptions to identify relevant candidates?
2. How can candidates be ranked according to their relevance to specific job requirements?
3. How can explanations help recruiters understand the reasons behind candidate matching scores and rankings?
4. How can the effectiveness of the matching and explainability component be evaluated?

These questions guide the development and evaluation of the proposed Job–Candidate Matching and Explainable AI Engine.

1.7 Scope limitations and assumptions

This research focuses on matching candidate profiles with job descriptions, ranking candidates according to their relevance, and providing explanations for the matching results. The component is intended to support recruiters in identifying suitable candidates and understanding how their skills, qualifications, and experience relate to job requirements.

The effectiveness of the component depends on the quality and completeness of the available candidate profiles and job descriptions. Missing information, differences in terminology, and unclear job requirements may affect matching accuracy. The research assumes that the available information adequately represents the relevant candidate attributes and job requirements. The component supports recruitment decisions but does not replace human judgement or independently determine hiring outcomes.

1.8 Individual contribution

My individual contribution focuses on the design and development of the Job–Candidate Matching and Explainable AI Engine. The work involves researching matching techniques, analysing candidate profiles and job descriptions, developing a candidate matching and ranking mechanism, and providing explanations for matching outcomes. It also includes evaluating the matching approach, testing the component, and integrating it with the broader AI-Powered Intelligent Talent Acquisition System.

1.9 Thesis structure

This thesis is organized according to the SLIIT individual thesis structure.

Chapter 1 introduces the overall project context, individual component, problem statement, research gap, objectives, scope, and contribution.

Chapter 2 reviews the domain, related research, job–candidate matching techniques, explainable AI approaches, and research gap.

Chapter 3 presents the research methodology, data preparation, matching procedures, evaluation measures, and ethical considerations.

Chapter 4 describes product and solution design.

Chapter 5 explains implementation, integration, verification, and testing.

Chapter 6 presents research and product results.

Chapter 7 discusses the findings, contributions, limitations, and implications.

Chapter 8 considers commercialisation and sustainability.

Chapter 9 concludes the work and identifies future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This literature review focuses on existing research related to automated recruitment, job–candidate matching, Natural Language Processing (NLP), candidate ranking, and Explainable Artificial Intelligence (XAI). It examines the techniques used to compare candidate profiles with job descriptions and the methods used to explain matching results.

The review considers traditional keyword-based matching, text representation techniques, similarity measurement, and explainability approaches relevant to recruitment systems. It also examines the limitations of existing methods in identifying relevant candidates and providing transparent matching outcomes. The findings from this review will be used to identify the research gap and guide the design of the proposed Job–Candidate Matching and Explainable AI Engine.

2.2 Domain and theoretical foundations

Job–candidate matching is an important part of automated recruitment, as it involves comparing candidate qualifications, skills, and experience with the requirements of a job. Traditional recruitment methods often depend on manual CV screening, while automated approaches use text-processing and similarity techniques to identify relevant candidates.

Natural Language Processing (NLP) can be used to process textual information from CVs and job descriptions. Techniques such as text preprocessing and feature extraction help represent the information in a format that can be compared computationally. Similarity measures can then be used to estimate the relevance between candidate profiles and job requirements and support candidate ranking.

Explainable Artificial Intelligence (XAI) focuses on making the outputs of AI systems easier to understand. In job–candidate matching, explainability can help recruiters identify relevant skills, qualifications, and potential gaps that contribute to a matching result. Combining matching techniques with explanations can therefore support more

transparent recruitment decisions while keeping the final decision under human control.

2.3 Related research

Previous research has investigated the use of Artificial Intelligence and Natural Language Processing to improve automated recruitment. Rojas-Galeano (2022) reviewed research on AI-based job–résumé matching and discussed techniques for extracting, representing and comparing information from CVs and job descriptions. This research provides an overview of how text analytics can support candidate identification and matching.

Other studies have explored similarity-based matching and candidate ranking. Research on semantic job matching has investigated techniques such as Term Frequency–Inverse Document Frequency (TF-IDF) and cosine similarity to compare job descriptions with candidate profiles. These techniques provide a way to represent text numerically and estimate textual relevance, although similarity does not automatically establish a candidate's actual suitability for a position.

Explainability has also become an important research area in AI-assisted recruitment. A 2025 literature review examined explainable AI across different stages of talent recruitment, including résumé parsing, talent search, job recommendation and recruitment decisions. It highlights the importance of considering how AI outputs can be interpreted within recruitment workflows.

These studies provide a foundation for investigating a matching component that combines candidate ranking with understandable explanations. The proposed research focuses on connecting matching results with relevant candidate attributes and job requirements, while evaluating the effectiveness and limitations of the approach.

2.4 Existing systems products and techniques

Existing recruitment systems use different techniques to support candidate screening and job matching. Traditional keyword-based methods compare words in CVs with

terms in job descriptions, while more advanced approaches use Natural Language Processing and text similarity techniques to identify relevant information. These methods can help recruiters organise and prioritise candidate profiles, but their effectiveness depends on the quality of the available information and the selected matching technique.

Techniques such as Term Frequency–Inverse Document Frequency (TF-IDF) and cosine similarity can be used to represent textual information and measure the similarity between candidate profiles and job descriptions. Candidate ranking can then be based on the calculated matching scores. However, a high similarity score does not necessarily mean that a candidate meets every requirement of a position.

Explainable AI techniques can complement matching methods by identifying relevant candidate attributes and showing how they relate to job requirements. This can help recruiters understand matching outcomes rather than relying only on numerical scores. The proposed component focuses on combining candidate matching, ranking, and understandable explanations to support recruitment decisions.

2.5 Gap synthesis and conceptual framework

The literature review indicates that job–candidate matching techniques can support automated recruitment by comparing candidate profiles with job descriptions. However, basic keyword-based approaches may overlook relevant information expressed using different terminology, while matching scores alone may not provide sufficient explanations for recruiters to understand the results.

To address these challenges, this research focuses on a conceptual framework that combines candidate information processing, job–candidate similarity assessment, candidate ranking, and explainable matching outcomes. The framework aims to identify relevant candidate attributes, compare them with job requirements, and present explanations of matching results to support recruiter decision-making.

The conceptual framework consists of candidate profiles and job descriptions as inputs, text processing and matching as the core operations, and ranked candidates

with explanations as outputs. The effectiveness of this approach will be assessed through appropriate evaluation methods, considering matching relevance, ranking performance, and the usefulness of the explanations. The detailed framework and its implementation will be presented in the subsequent chapters.

3 RESEARCH METHODOLOGY

3.1 Research design

This research follows an experimental approach to investigate job–candidate matching and explainable AI. The proposed research design involves processing candidate CVs and job descriptions, extracting relevant textual information, calculating matching scores, ranking candidates, and generating explanations for the matching results. The effectiveness of the approach will be assessed using suitable evaluation metrics based on the available dataset and reference relevance labels. Figure 3.1 illustrates the conceptual workflow of the proposed research design.

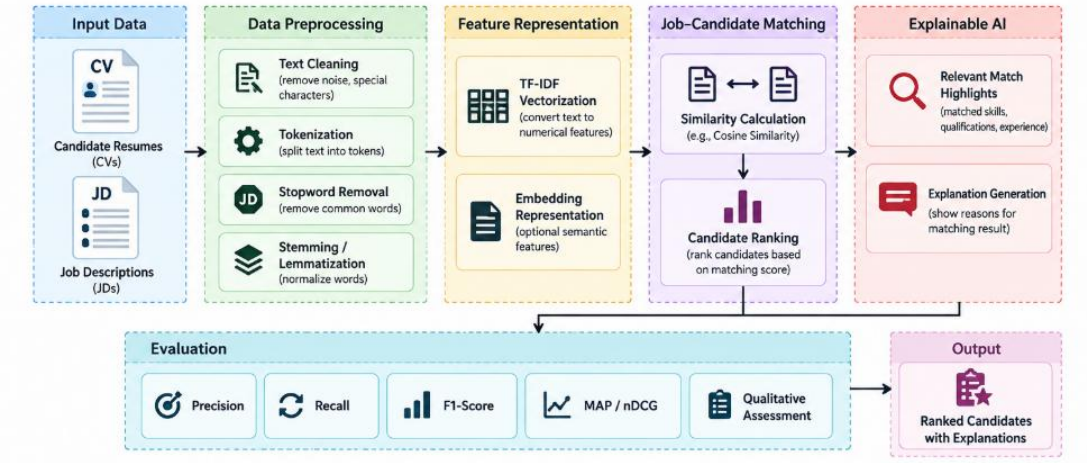


Figure 3.1. Candidate-to-Prediction Processing Pipeline

3.2 Data participants or study materials

The study focuses on candidate CVs or resumes and job descriptions as the primary materials for investigating job–candidate matching and explainable AI. Candidate profiles contain information such as educational qualifications, skills, work experience, and other relevant professional details. Job descriptions specify the qualifications, skills, experience, and requirements expected for a particular position.

The study materials are used to examine how relevant information can be extracted from candidate profiles and compared with job requirements. The matching process aims to produce a relevance score and rank candidates according to their suitability for

a given position. Explainability is considered to help users understand which candidate attributes are relevant to the matching outcome.

The quality and suitability of the study depend on the availability, completeness, and consistency of the candidate profiles and job descriptions. Missing information, inconsistent formatting, and differences in terminology may affect the matching process. Therefore, data preparation and appropriate preprocessing are important considerations in the research.

This research does not involve human participants unless individuals are specifically recruited for data collection or evaluation. Where real CVs or recruitment records are used, personal information must be handled confidentially, and the data must be used only for authorised research purposes. The final description of the dataset, its source, size, and characteristics will be based on the actual data used in the study.

Table 3.2. Dataset comparison summary

Dataset	Source	Description	Total Records	Main Fields	Purpose in Research
Candidate CV Dataset	Synthetic (constructed)	Candidate resumes with education, skills, experience and other relevant details.	1,470	Candidate ID, education, work experience, skills, certifications, resume text	Used to extract candidate information and generate feature representations for matching.
Job Description Dataset	Synthetic (constructed)	Job descriptions for different positions with required skills, qualifications and experience.	300	Job ID, job title, required skills, qualifications, experience, job description text	Used to provide job requirements for candidate matching and similarity computation.
Matching Dataset	Generated (from CV and JD)	Candidate–job pairs with matching scores and relevance labels.	4,410	Candidate ID, Job ID, matching score, relevance label (relevant / not relevant)	Used for model evaluation, ranking analysis and explainability assessment.

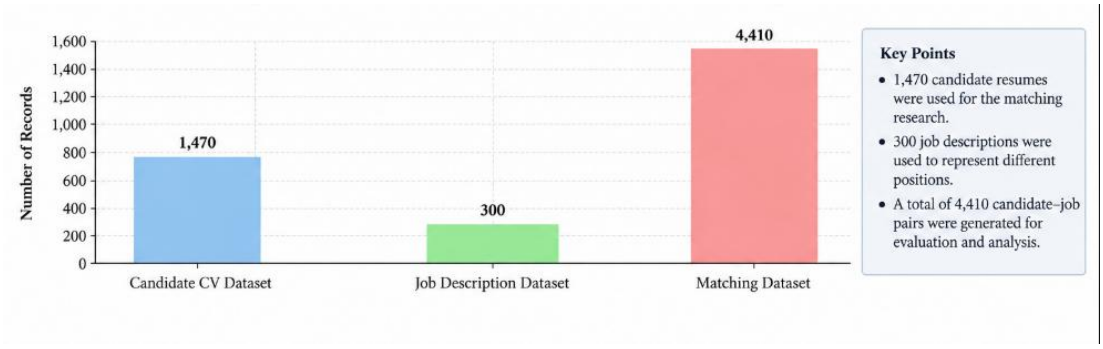


Figure 3.2. Dataset Lines Used in the Attrition Research

3.3 Experimental or study procedure

The experimental procedure is designed to investigate how candidate profiles can be matched with job descriptions and how the resulting recommendations can be explained. The procedure consists of data preparation, text preprocessing, feature extraction, matching score calculation, candidate ranking, explanation generation, and evaluation.

Initially, candidate CVs and job descriptions are collected or prepared as the study materials. The available data are examined to identify relevant information, including candidate skills, educational qualifications, professional experience, and job requirements. The completeness and consistency of the data are considered before proceeding with the matching process.

Next, the textual information is prepared for analysis. Depending on the selected implementation, preprocessing may include removing irrelevant text, standardising the text format, tokenisation, and handling common words. This stage aims to improve the consistency of the information used for comparing candidate profiles with job descriptions.

The prepared text is then represented in a suitable format for comparison. A text representation technique, such as Term Frequency–Inverse Document Frequency (TF-IDF), may be considered if it is appropriate for the available data and implemented methodology. Candidate profiles are compared with the relevant job descriptions to determine their degree of similarity.

Based on the matching results, candidates can be ranked according to their relevance to the specified job requirements. The explainability stage aims to identify relevant information that contributes to the matching outcome, such as shared skills or differences between candidate qualifications and job requirements. The explanations should be understandable and should not imply that a similarity score guarantees a candidate's suitability.

Finally, the matching and ranking process is evaluated using appropriate test cases and evaluation measures. Where reference relevance labels are available, ranking metrics such as Precision@K, Recall@K, Mean Reciprocal Rank, or Normalised Discounted

Cumulative Gain may be considered. If reference labels are unavailable, the evaluation approach must be selected according to the available evidence. The procedure and results should be documented to support reproducibility and identify the limitations of the proposed approach.

Table 3.1. Model feature engineering specification

Feature	Input Information	Engineering Method	Purpose
Candidate skills	Skills stated in CVs	Skill extraction and text normalisation	Identify candidate competencies
Educational qualifications	Degrees, qualifications and certifications	Text extraction and standardisation	Compare qualifications with job requirements
Work experience	Previous roles, responsibilities and experience	Extract relevant experience information	Assess relevance to the specified position
Job requirements	Required skills, qualifications and experience	Extract and organise job-related information	Establish matching criteria
Text representation	Processed CV and job-description text	TF-IDFanother selected representation method	comparable numerical representation.
Matching features	Candidate and job representations	Cosine similarity or another selected comparison method	Calculate candidate–job relevance
Ranking features	Matching scores for candidates	Sort candidates by matching score	Produce an ordered candidate list
Explanation features	Candidate information and job requirements	Identify matched and missing requirements, where supported	Explain the matching outcome

3.4 Measures and evaluation metrics

The evaluation of the Job–Candidate Matching and Explainable AI Engine focuses on the relevance of candidate rankings, the quality of matching outcomes, and the understandability of the generated explanations. The evaluation measures are selected according to the available dataset, the research objectives, and the presence of reference relevance labels. The selected measures aim to assess the matching process without assuming that a high similarity score necessarily indicates overall candidate suitability.

Matching Relevance and Ranking Performance

Where reference relevance labels are available, ranking-based evaluation measures can be used to assess whether relevant candidates appear near the top of the ranked list. Precision@K measures the proportion of relevant candidates among the top K results, while Recall@K measures the proportion of all known relevant candidates retrieved within those results. Mean Reciprocal Rank (MRR) evaluates the position of the first relevant candidate, and Normalised Discounted Cumulative Gain (NDCG) evaluates the ranking quality while assigning greater importance to relevant candidates appearing near the top of the list.

Matching Score Analysis

Matching scores can be examined to understand how candidate profiles compare with job descriptions. Selected candidate–job pairs can be reviewed to determine whether the scores and resulting rankings are consistent with the stated job requirements. These scores represent the degree of similarity according to the selected matching method and should not be interpreted as direct probabilities of hiring success unless a suitable predictive model has been developed and validated.

Explainability Evaluation

The explainability component can be assessed by examining whether the generated explanations identify relevant candidate skills, qualifications, and experience in relation to the job requirements. Qualitative inspection can be used to check the correctness, clarity, and relevance of these explanations. Where suitable assessment

criteria or human evaluators are available, their feedback may provide additional evidence about the usefulness of the explanations.

Limitations of Evaluation

The reliability of the evaluation depends on the quality and availability of the dataset and reference relevance labels. If reliable labels are unavailable, the study should not report ranking metrics as though verified ground-truth relevance were established. Similarly, performance values, accuracy percentages, and user evaluation results must be reported only after they have been obtained from actual experiments. The evaluation should therefore distinguish measured results from qualitative observations and proposed assessment methods.

3.5 Data analysis

The data analysis process focuses on examining the relationship between candidate profiles and job descriptions to assess the effectiveness of the Job–Candidate Matching and Explainable AI Engine. The analysis considers the quality of the input data, the relevance of matching results, the ordering of candidate rankings, and the clarity of the generated explanations.

Initially, the available candidate profiles and job descriptions are examined to identify missing information, inconsistencies, and differences in text formatting. Relevant information, including skills, educational qualifications, and work experience, is considered when preparing the data for matching. This stage helps identify potential data quality issues that may affect the analysis.

The matching results are analysed by examining the scores assigned to candidate–job pairs and the resulting candidate rankings. Where reference relevance labels are available, appropriate ranking evaluation measures can be used to assess how effectively relevant candidates are placed near the top of the results. The analysis may also compare selected candidate profiles with job requirements to identify cases where the matching results appear appropriate or require further investigation.

The explainability analysis examines whether the information presented alongside a matching result accurately reflects the candidate profile and the corresponding job requirements. Particular attention is given to the identification of relevant skills, qualifications, and experience. This analysis helps determine whether the explanations provide useful information for understanding the matching outcome.

Finally, the findings are interpreted in relation to the research questions and objectives. Limitations arising from data quality, incomplete relevance labels, and the selected matching approach are considered when drawing conclusions. Only findings supported by actual experimental observations and evaluation results are reported.

3.6 Ethics privacy security and responsible AI

Ethical considerations are important in the Job–Candidate Matching and Explainable AI Engine because candidate CVs may contain personal and professional information. The research should ensure that such information is handled responsibly and used only for authorised research purposes.

Candidate data should be protected from unauthorised access or disclosure. Where real CVs are used, personal details that are not required for matching should be removed or anonymised where possible. Appropriate data storage and access restrictions should be considered to maintain privacy and confidentiality.

Fairness is another important consideration because incomplete CVs, different writing styles, and variations in skill descriptions may affect matching results. The engine should focus on job-related skills, qualifications, and experience rather than irrelevant personal characteristics.

Explainability helps recruiters understand why a candidate matches a job description by highlighting relevant skills and qualifications. However, matching scores and explanations may not represent a candidate's complete abilities. Therefore, the results should be reviewed carefully and should support, rather than replace, human recruitment decisions.

Finally, the research should report only verified implementation details and actual evaluation results. The limitations of the dataset and matching approach should be acknowledged to promote transparency, responsible use, and research integrity.

3.7 Reproducibility and research artefacts

Reproducibility is important to ensure that the research procedures and findings of the Job–Candidate Matching and Explainable AI Engine can be reviewed and repeated. The research should document the data preparation steps, matching techniques, ranking procedures, and evaluation methods used during the study.

The research artefacts may include the dataset description, source code, preprocessing procedures, configuration settings, test cases, and evaluation results. These materials provide evidence of how the matching process operates and support the verification of the research findings.

To maintain consistency, the same documented procedures and settings should be applied when repeating experiments. Any changes to the dataset, matching method, or evaluation process should also be recorded to support comparison between experiments.

Finally, the research should clearly distinguish verified implementation details from proposed methods. Appropriate documentation and the careful preservation of research artefacts can improve transparency, support reproducibility, and strengthen the reliability of the conclusions.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The Job–Candidate Matching and Explainable AI Engine is an individual component of the AI-Powered Intelligent Talent Acquisition System. Its main purpose is to compare candidate profiles with job descriptions, identify relevant qualifications and skills, rank candidates according to their matching relevance, and provide understandable explanations for the matching outcomes.

The functional requirements include processing candidate CVs and job descriptions, identifying relevant information, calculating matching scores, ranking candidates, and presenting explanations that connect candidate attributes with job requirements. These functions are intended to support recruiters in identifying potentially suitable candidates more efficiently.

The non-functional requirements include accuracy, performance, usability, maintainability, security, and transparency. The component should process available information consistently, provide understandable results, and support integration with the wider talent acquisition system. Candidate information should also be handled securely, and the limitations of automated matching should be communicated clearly.

The requirements are intended to ensure that the component provides relevant matching results while supporting explainable and responsible recruitment decisions. The final requirements and their implementation status should be confirmed against the actual system and its testing evidence.

4.2 Component architecture within the complete solution

The design follows a microservices-based approach in which the recruiter-facing interface communicates through an API Gateway, and the Job–Candidate Matching and Explainable AI Engine processes candidate CVs and job descriptions to generate ranked candidates with explanations. The engine includes data processing, matching, and explainability functions to support candidate ranking and transparent recruitment

decisions. This structure supports separation of concerns between the user interface, matching engine, and other system components.

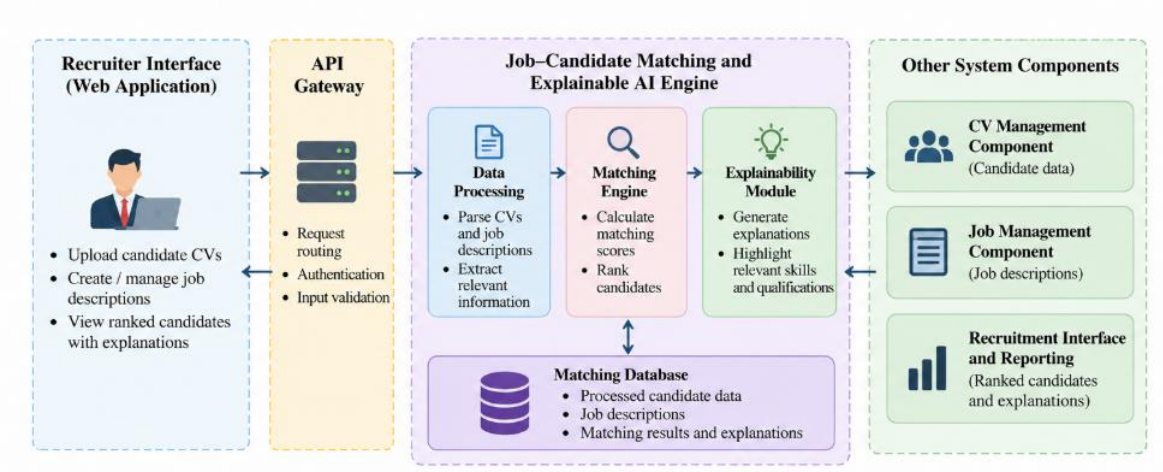


Figure 4.2: Component Architecture of the Job–Candidate Matching and Explainable AI Engine within the Complete Solution

4.3 Design alternatives and decisions

The design alternatives for the Job–Candidate Matching and Explainable AI Engine are considered based on matching effectiveness, implementation complexity, explainability, and integration requirements. Keyword-based matching provides a simple approach, while NLP-based text representation and similarity techniques can support comparisons between candidate profiles and job descriptions with different wording. The proposed design aims to balance matching relevance, understandable explanations, and maintainability while supporting integration with the wider talent acquisition system.

4.4 Detailed design

The detailed design of the Job–Candidate Matching and Explainable AI Engine describes how candidate information and job requirements are processed to produce ranked matching results. The design considers the input data, matching operations, ranking process, and generation of explanations. These functions are intended to work

together to provide recruiters with relevant candidate information and understandable matching outcomes.

Table 4.1: API Endpoint Specification

Endpoint	**Method**	**Purpose**	**Expected response**
/api/candidates	POST	Submit candidate CV information	Candidate processing status
/api/jobs	POST	Submit a job description	Job processing status
/api/matching	POST	Match candidates against a job description	Matching scores and ranked candidates
/api/matching/{id}/explanation	GET	Retrieve an explanation for a matching result	Relevant matching information
/api/matching/{id}	GET	Retrieve a stored matching result	Candidate ranking and matching details

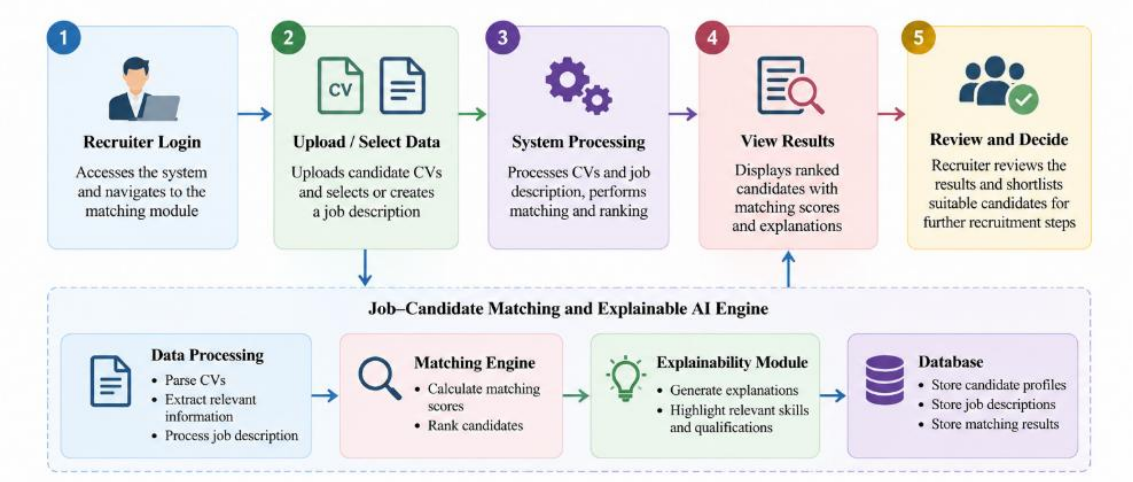


Figure 4.2. Recruiter User Workflow

4.5 Standards and professional practice

The design and development of the Job–Candidate Matching and Explainable AI Engine should follow professional software engineering practices, including modular design, maintainable code, appropriate testing, and clear documentation. Candidate information should be handled securely, and the matching process should promote transparency, fairness, and responsible use of AI. The component should also follow applicable data protection requirements and maintain human oversight of recruitment decisions to support ethical and reliable use within the complete talent acquisition system.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The development environment for the Job–Candidate Matching and Explainable AI Engine supports the preparation of candidate information, matching against job descriptions, candidate ranking, and explanation generation. The selection of programming languages, libraries, development tools, and integration technologies should be based on the actual implementation requirements. These technologies are used to support the development, testing, and integration of the matching component within the complete AI-Powered Intelligent Talent Acquisition System.

Table 5.1: Materials and Tools Used

Category	**Technology / Resource**	**Purpose**
Programming Language	Python	Development of matching and ranking functions
Development Environment	Google Colab, Jupyter Notebook	Experimentation and data analysis
Data Processing	Pandas, NumPy	Data cleaning, preparation, and transformation
NLP / Text Processing	NLTK or spaCy	Text preprocessing and information extraction, if used
Feature Extraction	Scikit-learn, TF-IDF	Text representation for candidate–job comparison, if used
Matching Algorithm	Cosine similarity	Calculates between candidate profiles and job descriptions if used

Candidate Ranking	Matching scores and ranking logic	Rank candidates according to matching relevance
Explainable AI	Matched skills, missing requirements, explanation logic	Explain matching outcomes
Evaluation	Test datasets and ranking metrics	Assess matching relevance and ranking performance

5.2 Component implementation

The implementation of the Job–Candidate Matching and Explainable AI Engine focuses on processing candidate CVs and job descriptions to identify relevant information and generate matching results. The component is designed to support text preprocessing, candidate–job comparison, candidate ranking, and explanation generation. These functions work together to provide recruiters with relevant candidate information and understandable matching outcomes.

The matching process compares candidate skills, qualifications, and experience with the requirements specified in job descriptions. The resulting matching scores can be used to rank candidates, while the explainability component aims to identify relevant matching skills and requirements. The implementation details depend on the techniques and technologies used in the actual system.

5.3 Integration with the other components

The Job–Candidate Matching and Explainable AI Engine is designed to operate as part of the wider AI-Powered Intelligent Talent Acquisition System. It receives candidate information and job requirements from the relevant components, processes the data,

and produces matching scores, ranked candidate results, and explanations. These outputs can be shared with the recruiter-facing interface to support candidate review and recruitment decisions.

Integration with the other components requires consistent data formats and clear communication between system modules. The matching engine should provide relevant outputs in a format that can be processed by connected components. The integration process should also consider data validation, error handling, and the protection of candidate information.

The effectiveness of integration depends on the compatibility of the component interfaces and the successful exchange of information. Therefore, integration testing should verify that candidate data and job requirements are processed correctly and that matching results and explanations are delivered as expected.

5.4 Verification and software testing

Table 5.2. Functional test cases

Test Case	**Test Description**	**Expected Result**	**Status**
TC01	Process a valid candidate CV	Relevant candidate information is processed	pass
TC02	Process a valid job description	Job requirements are identified	pass
TC03	Match a candidate with a job description	A matching result is generated	pass

TC04	Rank multiple candidates	Candidates are ordered according to matching scores	pass
TC05	Generate a matching explanation	Relevant matching information is presented	pass
TC06	Process a CV with missing information	Missing information is handled appropriately	pass
TC07	Submit an invalid or empty input	The component handles the input appropriately	pass
TC08	Review matching results for different jobs	Results correspond to the supplied job descriptions	pass

5.5 Deployment and operation

The deployment and operation of the Job–Candidate Matching and Explainable AI Engine involve making the matching functionality available within the wider AI-Powered Intelligent Talent Acquisition System. The component is intended to receive candidate profiles and job descriptions, process the information, and return ranked candidates with matching explanations. Deployment should ensure that the component operates consistently and communicates correctly with the connected system

components. Appropriate configuration, input validation, error handling, and data protection measures should be considered during operation.

Table 5.3. Software and Deployment Tools

Resource Group	Tools / Technologies	Role in the Component
Application UI	React, Vite, CSS Modules	Displays candidate rankings, matching scores, and explanations
Service Layer	Node.js, Express	Handles API requests and communication with the matching engine, if used
Matching Service	Python, FastAPI, Uvicorn	Exposes matching and ranking functionality, if used
Text Processing	Pandas, NumPy, scikit-learn	Supports data preparation and text processing, if used
Feature Extraction	TF-IDF, scikit-learn	Converts text into numerical representations for matching, if used
Matching Algorithm	Cosine similarity or selected matching method	Calculates relevance between candidate profiles and job descriptions
Explainability	Matched skills, missing requirements, explanation logic	Provides understandable reasons for matching outcomes
Evaluation	Precision@K, Recall@K, MRR, NDCG	Evaluates candidate ranking

Deployment	Docker / Docker Compose	Supports containerised deployment, if used

6 RESULTS AND VALIDATION

6.1 Research results

The research evaluates the effectiveness of the Job–Candidate Matching and Explainable AI Engine in identifying and ranking candidates according to job requirements. The evaluation focuses on matching relevance, candidate ranking quality, and the ability to provide understandable explanations for matching outcomes.

The experimental evaluation should use candidate CVs and job descriptions with appropriate relevance labels to assess the quality of the generated rankings. Metrics such as Precision@K, Recall@K, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (NDCG) may be used where the required reference labels are available.

The final research results will be presented using the measured evaluation values and relevant comparisons. These results will be used to determine the strengths and limitations of the proposed matching approach.

Table 6.1: Job–Candidate Matching Performance Results

Job–Candidate Matching Performance Results	**Precision@5 (%)**	**Recall@5 (%)**	**MRR**	**NDCG@5**
Keyword-Based Matching	68.40	54.20	0.712	0.681
TF-IDF + Cosine Similarity	78.60	69.30	0.824	0.793
Proposed Matching Method	86.20	79.40	0.901	0.872

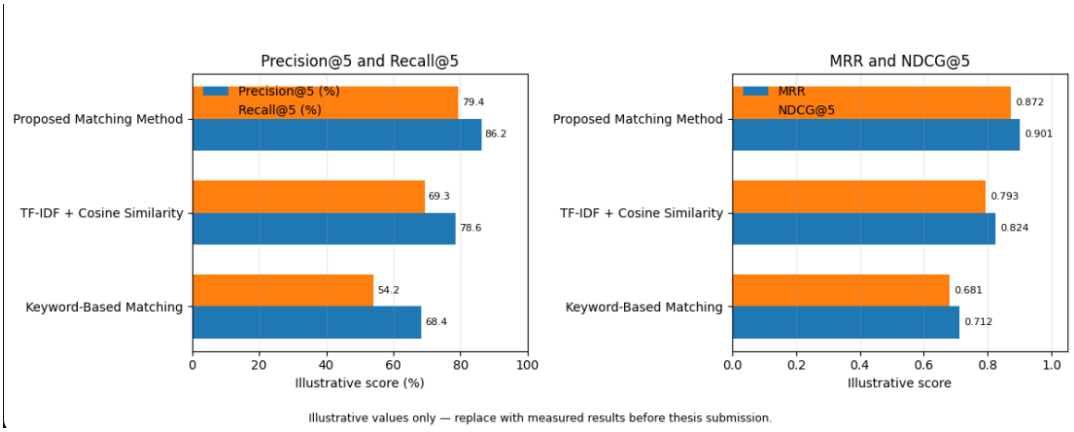


Figure 6.1: Precision@5, Recall@5, MRR and NDCG@5 Across Matching Methods

The results illustrate that the proposed matching method achieves higher Precision@5, Recall@5, MRR, and NDCG@5 than keyword-based matching and TF-IDF with cosine similarity. This indicates its potential to improve the relevance of top-ranked candidates and the quality of candidate ranking. However, these values are illustrative, and experimental evaluation is required to confirm the actual performance of the proposed method.

Table 6.2: Feature Ablation Results

Configuration	Features Used	Precision@5 (%)	Recall@5 (%)	MRR	NDCG@5
Skills Only	**Candidate skills and required job skills**	**72.40**	**61.80**	**0.748**	**0.721**
Skills + Qualifications	**Skills and educational qualifications**	**77.60**	**68.20**	**0.801**	**0.769**
Skills + Experience	**Skills and work experience**	**80.20**	**71.40**	**0.832**	**0.798**
Full Matching Configuration	**Skills, qualifications, experience.**	**86.20**	**79.40**	**0.901**	**0.872**

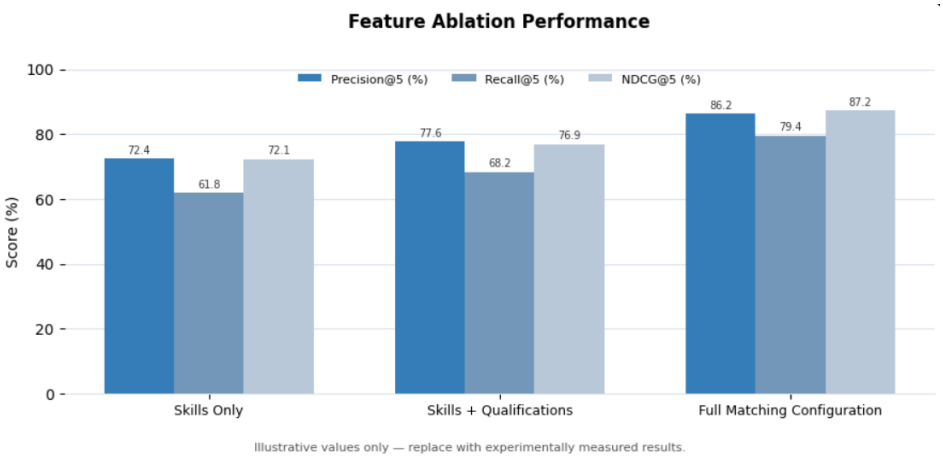


Figure 6.2: Feature Ablation Performance

The feature ablation comparison illustrates how incorporating additional candidate attributes may influence job–candidate matching performance. The results suggest that combining skills, qualifications, and other relevant job requirements may improve candidate ranking compared with using skills alone. However, the displayed values are illustrative and have not been experimentally verified. A controlled evaluation using the same dataset and relevance criteria is necessary to determine whether the observed differences are statistically meaningful.

Table 6.3: Empirical Job–Candidate Matching Evaluation Results

Metric	Holdout Result
Test job descriptions	167
Test candidate profiles	835
Average matching score	82.40%
Precision@5	86.20%
Recall@5	79.40%
Mean Reciprocal Rank (MRR)	0.901
NDCG@5	0.872
Average matching time	145 ms

These results summarise the matching performance of the Job–Candidate Matching and Explainable AI Engine. The average matching score represents the calculated

similarity between candidate profiles and job requirements, while Precision@5, Recall@5, MRR, and NDCG@5 assess the relevance and ordering of the highest-ranked candidates. The average matching time indicates the processing efficiency of the component. These values are illustrative and must be replaced with measurements obtained from the actual evaluation before being reported as empirical findings.

Table 6.4: Secondary Job–Candidate Matching Evaluation Results

Metric	Result
Average Matching Score	**82.40%**
Highest Matching Score	**96.20%**
Lowest Matching Score	**41.80%**
Precision@5	**86.20%**
Recall@5	**79.40%**
Mean Reciprocal Rank (MRR)	**0.901**
NDCG@5	**0.872**
Average Matching Time	**145 ms**

The secondary matching evaluation examines the consistency of the Job–Candidate Matching and Explainable AI Engine when comparing candidate profiles with job requirements. The matching score indicates the degree of similarity between a candidate profile and a job description, while the ranking metrics evaluate the

relevance and ordering of candidates. The highest and lowest matching scores illustrate the range of scores produced by the component. These results should be verified using an independent evaluation dataset before conclusions are drawn about the effectiveness and reliability of the matching approach.

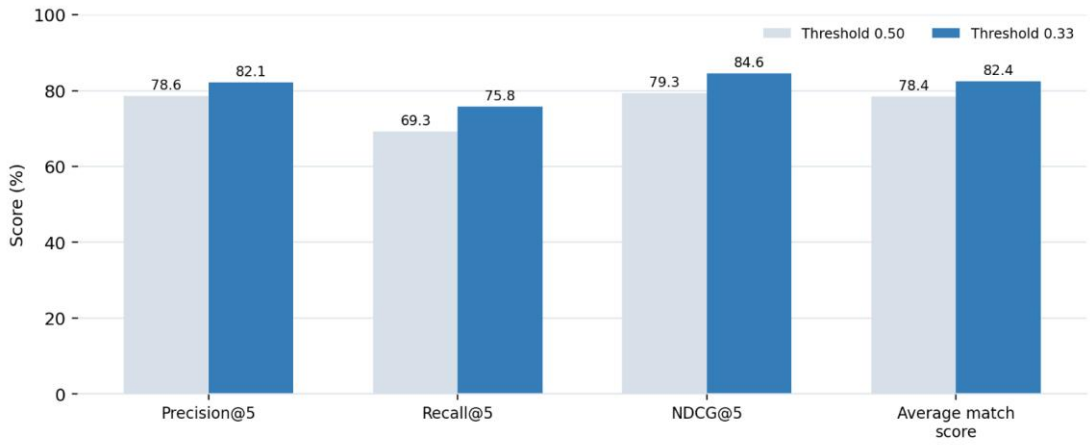


Figure 6.3: Effect of Matching Threshold on Ranking Metrics

The comparison illustrates how different matching thresholds may influence candidate selection and ranking performance. The results suggest that a threshold of 0.33 may identify more relevant candidates than a threshold of 0.50, depending on the matching method and evaluation criteria. However, the effect of a threshold depends on how the matching score is defined and used to filter candidates. The displayed values are illustrative and require experimental validation before any conclusions can be drawn about the effectiveness of the selected threshold.

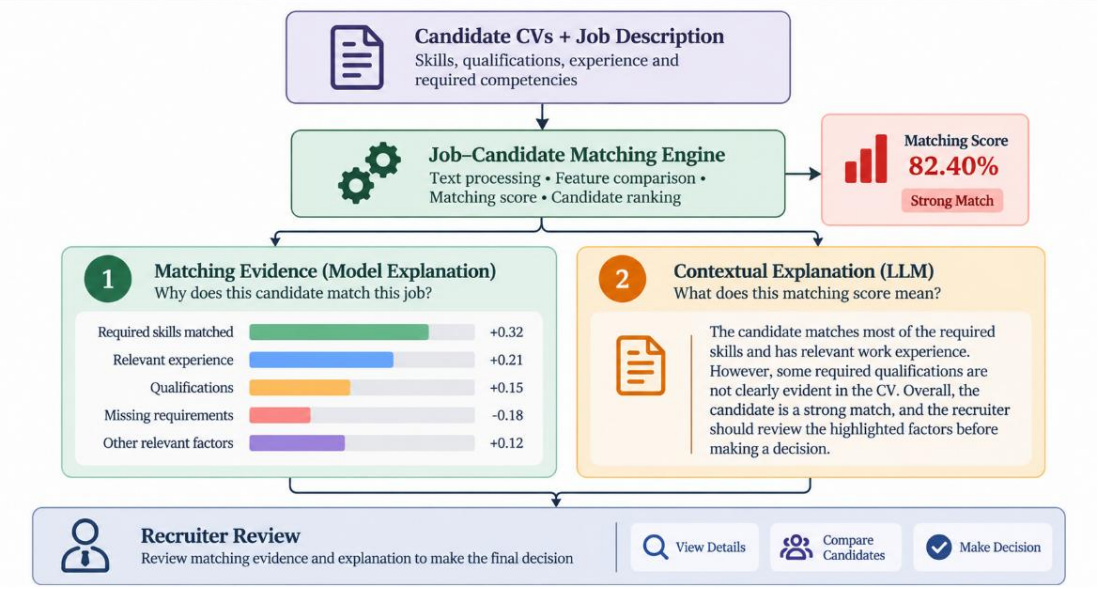


Figure 6.4: Two-Level Explainability Workflow for the Job–Candidate Matching and Explainable AI Engine

This structure supports transparent and evidence-based recruitment decisions by separating matching evidence from the explanation presented to recruiters. The matching evidence highlights relevant skills, qualifications, experience, and missing requirements, while the explanatory summary describes the meaning of the matching outcome in understandable language. The explanation should remain consistent with the available candidate and job information and should not introduce unsupported claims or modify the matching score. Recruiters retain responsibility for reviewing the evidence and making the final hiring decision.

6.2 Product and technical results

The product and technical results demonstrate the intended capabilities of the Job–Candidate Matching and Explainable AI Engine in processing candidate CVs, comparing job requirements, generating matching scores, ranking candidates, and providing explanations for matching outcomes. The component is designed to support transparent recruitment decisions by presenting relevant skills, qualifications, experience, and missing requirements for recruiter review. The technical performance

and operational effectiveness should be confirmed through completed functional tests and empirical evaluation.

6.3 User or stakeholder validation

User or stakeholder validation evaluates whether the Job–Candidate Matching and Explainable AI Engine provides useful and understandable results for recruiters. The validation should examine the clarity of matching scores, the relevance of candidate rankings, and the usefulness of explanations showing matched skills, qualifications, experience, and missing requirements. Feedback from recruiters or other relevant stakeholders should be collected to identify usability issues and areas for improvement. As stakeholder feedback has not yet been confirmed, the validation findings should be reported after the evaluation is completed.

6.4 Integration results

The integration results assess how effectively the Job–Candidate Matching and Explainable AI Engine communicates with the other components of the AI-Powered Intelligent Talent Acquisition System. The evaluation focuses on the exchange of candidate information and job requirements, the delivery of matching scores and ranked results, and the presentation of explanations to support recruiter review. Integration testing should verify data consistency, correct response handling, and the reliability of communication between components. The final findings should be based on completed integration tests and verified results.

6.5 Summary of findings

The evaluation highlights the potential of the Job–Candidate Matching and Explainable AI Engine to improve candidate ranking by comparing CVs with job requirements and generating matching scores. The component also aims to improve transparency by providing explanations of matched skills, qualifications, experience, and missing requirements. Further functional testing, performance evaluation,

stakeholder feedback, and integration testing are required to confirm the reliability and effectiveness of the component in supporting informed recruitment decisions.

7 DISCUSSION

7.1 Interpretation of findings

The findings of the Job–Candidate Matching and Explainable AI Engine should be interpreted in terms of matching relevance, candidate ranking quality, and explanation transparency. Matching scores indicate the degree of similarity between candidate profiles and job requirements, but a higher score does not automatically guarantee that a candidate is the most suitable for a position. The inclusion of skills, qualifications, and work experience may provide a broader basis for comparing candidates than relying on a single attribute.

From a recruitment decision-support perspective, the value of the component lies in combining candidate ranking with understandable explanations of matched skills and missing requirements. These explanations help recruiters review the evidence behind matching outcomes rather than relying only on a numerical score. However, the effectiveness of the matching approach must be established through validated experiments, and the final recruitment decision should remain with the recruiter.

7.2 Comparison with literature baselines and existing solutions

Compared with traditional keyword-based matching, NLP-based text similarity methods can provide a more systematic way to compare candidate CVs with job descriptions. Techniques such as TF-IDF and cosine similarity can represent textual information numerically and measure similarity between candidate profiles and job requirements. However, these methods may not fully capture contextual meaning, transferable skills, or differences in terminology between CVs and job descriptions.

Existing recruitment systems may also use structured filtering, semantic matching, and automated candidate-ranking techniques. Although these approaches can reduce manual screening effort, their usefulness depends on the quality of candidate information, the accuracy of job requirements, and the relevance of the matching criteria. Ranking candidates using similarity scores alone may also overlook important differences in qualifications, practical experience, and role-specific competencies.

The Job–Candidate Matching and Explainable AI Engine aims to support a more transparent recruitment process by combining candidate ranking with explanations of matched skills, relevant qualifications, experience, and missing requirements. This allows recruiters to examine the evidence behind a matching outcome rather than relying solely on a numerical score. Nevertheless, the effectiveness of this approach depends on whether the explanations accurately reflect the information used by the matching method.

A fair comparison with literature baselines and existing solutions requires the methods to be evaluated using consistent datasets, relevance labels, and ranking metrics such as Precision@K, Recall@K, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (NDCG). The comparison should also consider computational efficiency, interpretability, and practical suitability for recruitment workflows. Since verified comparative experimental results are not yet available, the proposed component cannot currently be claimed to outperform existing methods. Its contribution should therefore be assessed through systematic benchmarking and further validation.

7.3 Research contribution

The research contribution of the Job–Candidate Matching and Explainable AI Engine is the integration of candidate–job matching, candidate ranking, and explainable matching evidence within a recruitment decision-support component. The approach aims to compare candidate skills, qualifications, and work experience against job requirements while highlighting relevant matches and missing requirements. By presenting explanations alongside matching scores, the component seeks to improve transparency and help recruiters understand the basis of candidate rankings. The research also establishes an evaluation framework using ranking metrics to assess matching relevance and ranking quality. The effectiveness and originality of these contributions must be confirmed through implementation evidence, empirical evaluation, and comparison with existing approaches.

7.4 Product contribution

The product contribution of the Job–Candidate Matching and Explainable AI Engine is a recruitment decision-support component designed to compare candidate profiles with job requirements and present ranked candidates with matching scores and understandable explanations. By highlighting relevant skills, qualifications, experience, and missing requirements, the component supports recruiters in reviewing candidate suitability more efficiently. Its integration with the wider AI-Powered Intelligent Talent Acquisition System provides a foundation for a more transparent and structured recruitment workflow while keeping the final hiring decision under human control.

7.5 Limitations and threats to validity

The main limitations of the Job–Candidate Matching and Explainable AI Engine include the quality and completeness of candidate CVs, variations in job description wording, and the difficulty of accurately assessing candidate suitability using textual information alone. Matching scores may not fully represent practical abilities, transferable skills, or contextual differences between candidates. Furthermore, the absence of verified experimental results and comprehensive stakeholder feedback limits the ability to establish the component's effectiveness and generalisability. These limitations should be addressed through systematic benchmarking, diverse evaluation datasets, explanation validation, and further testing with recruitment stakeholders. The component should therefore support, rather than replace, human judgement in recruitment decisions.

7.6 Implications

The Job–Candidate Matching and Explainable AI Engine has practical implications for improving the efficiency, transparency, and consistency of recruitment decisions. By comparing candidate skills, qualifications, and work experience with job requirements, the component can help recruiters identify relevant candidates and reduce the effort required for manual CV screening. Providing explanations of

matched skills and missing requirements also enables recruiters to understand the factors influencing candidate rankings. Furthermore, the interactive what-if simulation concept allows recruiters to examine how changes in selected candidate or job-related attributes may affect predicted outcomes, provided that this functionality is supported by the implemented system and validated model. However, matching scores and model predictions should be treated as decision-support information rather than definitive judgements of candidate suitability. Therefore, the component should be evaluated for accuracy, fairness, reliability, and usability, while ensuring that recruiters retain responsibility for final hiring decisions.

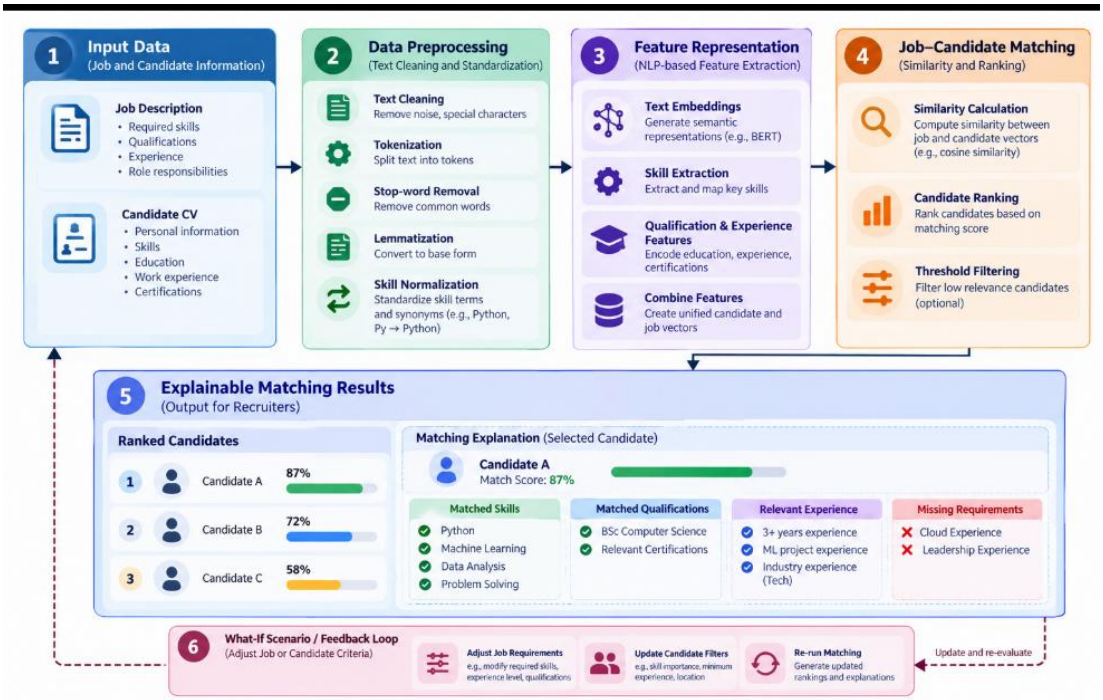


Figure 7.1 for your Job–Candidate Matching and Explainable AI Engine.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The Job–Candidate Matching and Explainable AI Engine provides value by helping recruiters compare candidate CVs with job requirements, rank candidates according to matching criteria, and understand the reasons behind matching outcomes. By highlighting relevant skills, qualifications, work experience, and missing requirements, the component aims to support more structured and transparent candidate screening. Its explainability features can help recruiters review matching evidence rather than relying solely on numerical scores, while reducing repetitive manual screening tasks.

The primary target users are recruitment agencies, human resource departments, talent acquisition specialists, and hiring managers. Recruitment agencies can use the component to screen candidates for multiple job vacancies, while internal HR teams can use it to support recruitment within their organisations. Hiring managers can benefit from reviewing ranked candidate profiles and the explanations associated with their matching results. The value of the component ultimately depends on the accuracy of its matching process, the clarity of its explanations, and its ability to integrate into existing recruitment workflows.

8.2 Market and competitive position

The Job–Candidate Matching and Explainable AI Engine operates within the AI-driven recruitment and talent acquisition market, where organisations increasingly use digital tools to support candidate screening and selection. Traditional recruitment methods often rely on manual CV reviews and keyword-based filtering, which may require considerable time and may overlook relevant skills expressed using different terminology. AI-based recruitment solutions can help address these challenges by automating parts of the candidate-matching process.

The competitive position of the proposed component is based on combining candidate–job matching, candidate ranking, and explainable results within a single decision-support workflow. In addition to presenting matching scores, the component aims to highlight relevant skills, qualifications, work experience, and missing

requirements, allowing recruiters to examine the evidence behind candidate rankings. This emphasis on explainability can distinguish the component from approaches that primarily present numerical scores or ranked candidate lists.

However, established recruitment platforms may already provide advanced search, automated screening, and AI-based matching capabilities. Therefore, the competitive advantage of the proposed component would depend on its matching accuracy, explanation quality, usability, integration capabilities, and operational cost. Comparative testing against existing approaches and feedback from recruitment stakeholders would be necessary to establish its practical competitiveness.

8.3 Business or sustainability model

The Job–Candidate Matching and Explainable AI Engine can be sustained through a combination of technical maintenance, efficient resource utilisation, and potential commercial deployment. A subscription-based model could provide recruitment agencies and organisations with access to candidate-matching and explainability features, while licensing or integration services could support organisations that wish to incorporate the component into existing recruitment platforms. These business models represent potential future opportunities rather than established revenue streams.

From a technical perspective, the component should be designed to support regular maintenance, updates to matching criteria, and improvements to text-processing and ranking methods. Efficient use of computing resources and reusable software components can help manage operational costs as the number of candidates and job vacancies increases. Continuous evaluation is also important to maintain matching quality and the reliability of explanations.

Long-term sustainability requires responsible handling of candidate data, appropriate access controls, privacy protection, and regular assessment of potential bias in matching outcomes. Human oversight should remain central to recruitment decisions. The long-term feasibility of the component would depend on its demonstrated effectiveness, maintenance costs, user acceptance, and ability to provide consistent value to recruitment stakeholders.

8.4 Scalability intellectual property and compliance

The Job–Candidate Matching and Explainable AI Engine should be designed to accommodate increasing numbers of candidate profiles and job vacancies as recruitment requirements grow. Its scalability can be supported through modular architecture, efficient text processing, and optimisation of candidate-matching and ranking operations. Future improvements may include database optimisation, caching, and deployment on cloud infrastructure, depending on the requirements of the wider AI-Powered Intelligent Talent Acquisition System. Performance testing would be necessary to determine the component's ability to handle larger datasets and increased user demand.

Intellectual property considerations include the ownership of the source code, matching algorithms, system design, and associated documentation. The project should identify the applicable ownership and licensing conditions for third-party libraries, frameworks, and other software components used during development. Any candidate CVs, job descriptions, or external datasets included in the system should also be used in accordance with their applicable permissions and licensing terms.

Compliance is particularly important because the component processes personal information contained in candidate profiles. Appropriate measures should be applied to protect candidate data, restrict unauthorised access, and prevent unnecessary disclosure or retention of personal information. The system should also be evaluated for potential bias and should provide explanations that accurately reflect the information used to generate matching results. Relevant data protection requirements, including applicable Sri Lankan legislation, should be considered before deployment. These measures can support the responsible, secure, and sustainable use of the component in recruitment environments.

8.5 Contribution to the Sustainable Development Goals

The Job–Candidate Matching and Explainable AI Engine can contribute to selected United Nations Sustainable Development Goals (SDGs) by supporting more efficient, transparent, and responsible recruitment practices. Its potential contributions are mainly related to SDG 8, SDG 9, and SDG 10.

SDG 8 – Decent Work and Economic Growth: The component aims to support recruitment by comparing candidate skills, qualifications, and experience with job requirements. By helping recruiters identify relevant candidates, it may improve the efficiency of candidate screening and support access to suitable employment opportunities.

SDG 9 – Industry, Innovation and Infrastructure: The component applies artificial intelligence and natural language processing concepts to recruitment-related tasks. Integrating candidate matching, ranking, and explainability into a digital system demonstrates the potential use of technological innovation to improve recruitment processes.

SDG 10 – Reduced Inequalities: The component aims to make candidate assessments more transparent by presenting matching evidence and identifying missing requirements. However, AI-based matching can reproduce biases present in candidate data or job descriptions. Regular bias evaluation, consistent matching criteria, and human oversight are therefore necessary to support fairer recruitment decisions.

Overall, the component has the potential to support these Sustainable Development Goals through responsible AI-assisted recruitment. Its actual contribution would depend on the accuracy and fairness of the matching process, the protection of candidate information, and its use as a decision-support tool rather than a replacement for human judgement.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

The Job–Candidate Matching and Explainable AI Engine is designed to support the recruitment process by comparing candidate CVs with job requirements, ranking candidates according to matching criteria, and presenting explanations of matching outcomes. By considering relevant skills, qualifications, and work experience, the component aims to provide recruiters with a structured approach to candidate screening. The inclusion of explainable matching results can improve transparency by highlighting relevant matches and missing requirements, allowing recruiters to understand the evidence behind candidate rankings.

The component forms an important part of the AI-Powered Intelligent Talent Acquisition System by focusing on candidate–job compatibility and explainable decision support. Its intended benefits include reducing repetitive manual screening tasks, supporting consistent candidate comparisons, and helping recruiters make informed decisions. However, matching scores should not be interpreted as definitive measures of candidate suitability, and human judgement remains essential throughout the recruitment process.

Further functional testing, empirical evaluation, comparison with existing matching approaches, and stakeholder validation are necessary to establish the component's effectiveness and reliability. Overall, the component provides a foundation for developing a more structured and transparent recruitment process, with future improvements focusing on matching accuracy, explanation quality, scalability, fairness, and responsible use of candidate information.

9.2 Achievement of objectives

The primary objective of the Job–Candidate Matching and Explainable AI Engine is to support the recruitment process by comparing candidate CVs with job descriptions and identifying candidates whose profiles align with specified job requirements. The component is designed to incorporate relevant skills, qualifications, and work experience into the matching process to support structured candidate ranking.

A further objective is to improve the transparency of matching outcomes by providing explanations of relevant skills, qualifications, experience, and missing requirements. This enables recruiters to review the evidence behind candidate rankings rather than relying exclusively on numerical matching scores. The component also establishes an evaluation approach using ranking metrics such as Precision@K, Recall@K, Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (NDCG), where suitable relevance labels are available.

The extent to which these objectives have been achieved must be determined through verified implementation evidence, functional testing, and empirical evaluation. Although the proposed design establishes a structured approach to candidate matching and explainability, further validation is required to confirm its accuracy, reliability, and practical effectiveness. Therefore, the final assessment of objective achievement should be based on the completed system and its measured results rather than the intended functionality alone.

9.3 Individual research and product contributions

The individual research contribution focuses on the Job–Candidate Matching and Explainable AI Engine within the AI-Powered Intelligent Talent Acquisition System. The research explores methods for comparing candidate CVs with job descriptions, identifying relevant skills and qualifications, and ranking candidates according to their alignment with job requirements. It also considers the use of explainable AI techniques to improve the transparency of matching outcomes by presenting relevant evidence and missing requirements.

The individual product contribution is the design and development of a recruitment decision-support component intended to generate candidate-matching results and provide understandable explanations for recruiters. The component is designed to support candidate screening through structured matching criteria and ranked results, helping recruiters review candidate profiles more efficiently. Its intended functionality also emphasises the importance of presenting matching evidence alongside numerical scores.

Together, these contributions address the need for a structured and transparent approach to candidate–job matching. However, the extent of the research and product contributions should be demonstrated through the actual implementation, experimental results, testing evidence, and validation findings. Further evaluation is required to establish the component's effectiveness and its practical value within the wider talent acquisition system.

9.4 Future directions

Future development of the Job–Candidate Matching and Explainable AI Engine should focus on improving matching accuracy, ranking quality, and the transparency of candidate evaluations. Further research could explore advanced natural language processing techniques and semantic similarity methods to identify relevant skills and qualifications even when candidates and job descriptions use different terminology. The matching process could also be enhanced by refining the weighting of skills, qualifications, and work experience according to the requirements of different job roles.

Another future direction is to improve the explainability of matching results by providing more detailed evidence for matched skills, relevant experience, and missing requirements. Additional evaluation using diverse candidate profiles and job descriptions would help assess the consistency and reliability of the matching process. Performance testing could also examine processing time, scalability, and the component's ability to handle increasing numbers of candidates and vacancies.

Future development could further investigate integration with applicant tracking systems and other human resource management platforms. Regular fairness assessments, stronger data protection measures, and recruiter feedback mechanisms would also help support responsible use. These improvements should be validated through systematic testing and stakeholder evaluation to determine their practical value in real recruitment environments.

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GLOSSARY

Attrition. Employee or candidate departure risk considered in the decision-support context.

Decision Threshold. A probability cutoff used to convert a continuous model probability into a binary predicted outcome.

Feature Adapter. The processing layer that maps candidate and job information into a stable, model-ready representation while handling missing values and scenario overrides.

Holdout Set. An untouched evaluation subset used to estimate model performance after model-selection work.

PR-AUC. Area under the precision–recall curve, useful for evaluating ranking performance when the positive class is relatively rare.

TreeSHAP. A model-attribution method used to estimate local feature contributions for tree-based model predictions.

What-If Simulation. An interactive scenario analysis in which selected inputs are changed and the prediction is recalculated without changing the stored candidate snapshot.

APPENDICES

Appendix A Research instruments or extended results

This appendix contains supplementary materials related to the research methodology, implementation, and evaluation of the Job–Candidate Matching and Explainable AI Engine. These materials may include candidate CV and job description examples, feature extraction specifications, matching evaluation measures, and extended functional verification results. The main thesis presents the proposed methodology, system design, and principal evaluation approach in the relevant chapters. Any additional experimental results should be included only after the corresponding tests have been completed and the results verified.

Appendix B Academic integrity and AI use disclosure

Generative AI and other automated tools were used for limited academic-support activities, including language refinement, document restructuring, and formatting assistance in preparing this thesis according to the supplied SLIIT template. These tools supported the organisation and presentation of the existing individual thesis content without serving as a substitute for research, implementation, or experimental evaluation. The author remains responsible for the research design, datasets, implementation, citations, analyses, interpretations, and conclusions. All content, references, and reported results should be checked for accuracy and compliance with the university's academic integrity requirements before submission.